

Homework week 3

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1 Question 1

For the following questions give both the output and its type. For example if the question is $a+b=c$, c is the output and its type is assumed to be number or integer. The questions are given in MATLAB code, so they should run, if they do not run then the error should be given as the answer.

1. 5^5
2. $16 + 30$
3. $32/4$
4. $25/2 * 6$

2 Question 2

Write MATLAB code for the following equations.

$$5 + 5 \quad (1)$$

$$\sqrt{4} \quad (2)$$

$$3^{1.2} \quad (3)$$

$$\tan 60 \quad (4)$$

3 Question 3

Store a variable *ftemp* giving it an initial value of 60. We will assume *ftemp* is the degrees of a room in Fahrenheit. Create a fxn to convert numbers from Fahrenheit to Celsius. Invoke your conversion fxn on four different variables including *ftemp*. Turn in your code as the answer.

4 Question 4

Write MATLAB expressions to generate a random number in the following ranges.

$$\begin{aligned}1 &\rightarrow 10 \\1 &\rightarrow 100 \\0 &\rightarrow 5 \\-100 &\rightarrow 100\end{aligned}$$

5 Question 5

Create the following matrix below.

$$M = \mathcal{R}^{10 \times 10} \tag{1}$$

6 Question 6

For the matrix *M* defined in question 8, do the following.

1. return the value that is at index $M_{1,1}$
2. return the value that is at index $M_{10,10}$
3. return the second row
4. return the third column

7 Question 7

Write the pseudocode for a function that would theoretically perform matrix multiplication.